The Data-Driven Social Engagement Initiative

Project Report | Social Media Analytics and Comment Sentiment Analysis

1. Introduction and Objective

Social media platforms generate large amounts of video and comment data. This project uses data analysis and basic natural language processing (NLP) to explore video performance, audience reactions, and content patterns. Its objective is to turn social engagement data into understandable insights that can support content planning and decision-making.

2. Dataset and Tools

The project uses YouTube trending-video data (USvideos.csv) and viewer comments (UScomments.csv). Python, Pandas, NumPy, Matplotlib, TextBlob, scikit-learn, and Streamlit are used for data cleaning, analysis, visualisation, sentiment scoring, basic modelling, and dashboard development.

3. Main Features Implemented

- Data loading and cleaning for video and comment datasets.
- Video performance summaries, including views, likes, comments, and engagement rate.
- Comment sentiment analysis using TextBlob polarity to assign Positive, Negative, or Neutral labels.
- A Streamlit dashboard with CSV upload, key metrics, tables, and charts.
- Content recommendations based on observed video data.
- Category comparison to explore differences in video performance.
- Historical views-trend visualisation using available trending-date records.
- A basic engagement classification experiment using scikit-learn.

4. Results and Limitations

The dashboard makes key video and comment metrics easier to explore and compare. Sentiment labels provide a simple overview of the tone of comments, while category and trend charts help identify patterns in the available historical data. The engagement model is a basic prototype and should not be treated as a reliable virality predictor without stronger features and evaluation. The Relatable-versus-Neutral comment classifier is not reported as completed because its manually labelled data still requires verification.

5. Conclusion and Future Work

This project demonstrates how Python-based analytics and a Streamlit dashboard can make social-media data more accessible. Future improvements include validating labels for relatable comments, evaluating models with suitable metrics and train/test splits, improving prediction features, and adding a structured content strategy report based on verified findings.

Note: Findings depend on the supplied datasets and the current prototype implementation. The project does not claim verified prediction accuracy for future virality.